

Experimental Setup

Environment

Model: Qwen2.5-VL-3B-Instruct

Platforms Used: Kaggle and Local PC

Hardware:

- NVIDIA RTX 3090 24GB VRAM (Local PC)

Training Dataset — Chart-RVR GRPO Train [15]

- Full dataset: 34,194 samples (ChartQA, PlotQA)
- Subset used: 1,000 samples - random sampling

Test Dataset — Chart-RVR GRPO Test [15]

- Total Samples Evaluated — 500
- In Distribution: ChartQA, PlotQA
- Out of Distribution: ChartFC

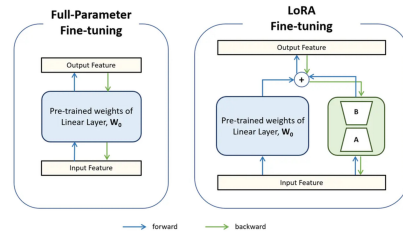
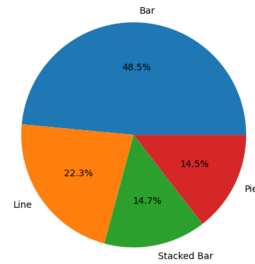


Chart Type Distribution (Pie Chart)



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Experimental Setup

Dataset Samples

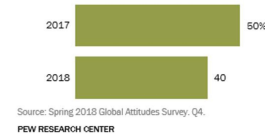
System Prompt

You are given a chart image and a query about the chart.

Think step-by-step about how to answer the query based on the chart image and then provide the final answer..

Decreasing Japanese satisfaction with functioning of democracy at home

Satisfied with the way democracy is working in Japan



Query

The difference between 2017 and 2018 in the percentage of Japanese satisfaction with democracy is what percentage?

Ground Truth : 10

Output Format

`<type> Bar </type>`

`<table>`

```
{
  "columns": ["Year", "Percentage"],
  "rows": [
    ["2017", "50%"],
    ["2018", "40%"]
  ]
}
```

`</table>`

`<think>`

`<step-1>: The question asks for ...`

`<step-2>: From the chart: 2017 = 50%, 2018 = 40%.`

`<step-3>: We need to calculate the difference.`

`<step-4>: The difference is 50% - 40% = 10%.`

`<step-5>: Values taken directly from chart.`

`Verified.`

`</think>`

`<answer> 10 </answer>`

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Results

Performance Evaluation — Experiment List

Exp. ID	Experiment	Method(s)	Purpose
E0	Base model evaluation	Qwen2.5-VL-3B-Instruct	Establish zero-training baseline
E1	Standard RLVR baseline	GRPO	Test whether flat verifiable rewards improve chart QA
E2	Proposed HCPC reward	GRPO+HCPC	Test hierarchical correct-path consistency vs flat GRPO
E3	Negative-only reinforcement baseline	NSR	Test whether text-only NSR transfers to VLM chart reasoning and compare with GRPO-HCPC
E4	OOD fact-checking transfer	GRPO, GRPO+HCPC, NSR	Test distribution shift from ChartQA-style QA to chart fact-checking

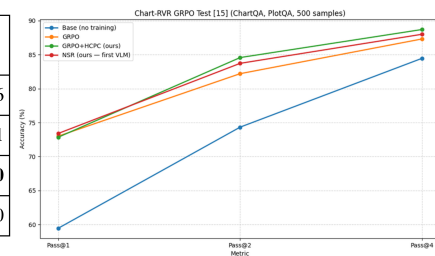
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Results

Performance Evaluation

Chart-RVR GRPO Test [15] (ChartQA, PlotQA, 500 samples)

Method	Pass@1	Pass@2	Pass@4
Base (no training)	59.45	74.33	84.46
GRPO	73	82.2	87.31
GRPO+HCPC (ours)	72.85	84.57	88.70
NSR (ours — first VLM)	73.4	83.72	88.00

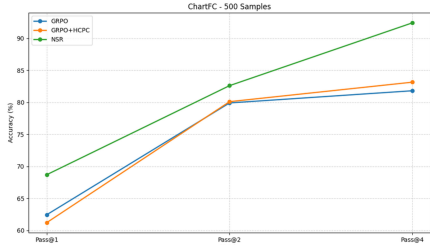


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Results
Performance Evaluation

ChartFC - 500 Samples

Table with 4 columns: Method, Pass@1, Pass@2, Pass@4. Rows include GRPO, GRPO+HCPC, and NSR.



References

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Conclusion and Future Plan

What We Did

- **HCPC (Hierarchical Correct-Path Consistency):** A cross-rollout reward bonus that rewards consistent table extraction and diverse reasoning — conditioned on correctness of intermediate steps, not just final answers
- **NSR applied to VLMs for the first time:** Extends negative-only reinforcement [32] to vision-language models with noisy, multi-component rewards. Used here as a competitive baseline to test whether implicit diversity preservation (withholding gradients on correct rollouts) can match explicit diversity engineering (HCPC).

Limitations &Future Plans

- **Model scale and recency** —Extending to larger parameter regimes (7B, 72B), Conducting experiments with K = 8.
- **Full training corpus** — Current results are based on a 1,000-sample subset of the 34,194-sample Chart-RVR dataset.
- **Implement CLC and NSR-HCPC Pipelines** — Add CLC reward and NSR-HCPC reward in pipeline to enhance reasoning robustness, consistency, and alignment with desired outputs.
- **Entropy-based reasoning diversity** — A principled alternative is to directly regularize the entropy of the rollout distribution during training, enabling explicit, continuous control over diversity without dependence on surface-level textual overlap.
- **Out-of-distribution robustness** — Investigate domain-adaptive training strategies and evaluate across a broader range of chart distributions.

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